Artificial Intelligence (AI)

Lecture - 01

Presented by

Rifa Zumana

Lecturer

Dept. Of CSE

North Bengal International University

Course Introduction

Couse No.: CSE 3111

Course Title: Artificial Intelligence

Contact Hours/Week: 3 hours

Credit Hour: 3

Couse No.: CSE 3112

Course Title: Artificial Intelligence Sessional

Contact Hours/Week: 2 hours

Credit Hour: 1

Course Outline

- History Of AI
- Intelligent Agents
- Problem Solving & Search
- Knowledge Representation
- Reasoning Under Uncertainty
- Planning And Learning
- AI Programming Languages (Prolog)
- Selected Topics (Neural Networks, Expert Systems, Robotics, Fuzzy Logic)

What is Artificial Intelligence (AI)?

- Artificial Intelligence (AI) is an area of computer science that emphasizes the creation of intelligent machines that work and react like humans.
- Artificial intelligence is complex in nature. It uses very complicated mixture of computer science, mathematics and other complex sciences.
- Complex programming helps these machines to replicate the cognitive abilities of human beings.

Can Machines Be Intelligent?

- Up to the present day it is not sure whether it is possible to build a machine that has all aspects of intelligence.

- This kind of research is central in the field of AI

Why Study AI?

- AI is everywhere: google maps, netflix, chatgpt, tesla
- Solves real-world problems (medicine, business, automation)
- Drives innovation in robotics, self-driving cars, and virtual assistants
- Helps understand intelligence itself

Goals of AI?

- As a science, AI attempts to establish a theory of intelligence to explain human intellectual activities and abilities.
- As a technology, AI attempts to implement a theory of intelligence in computer systems to reproduce these activities and abilities and use them to solve practical problems.

History Of AI

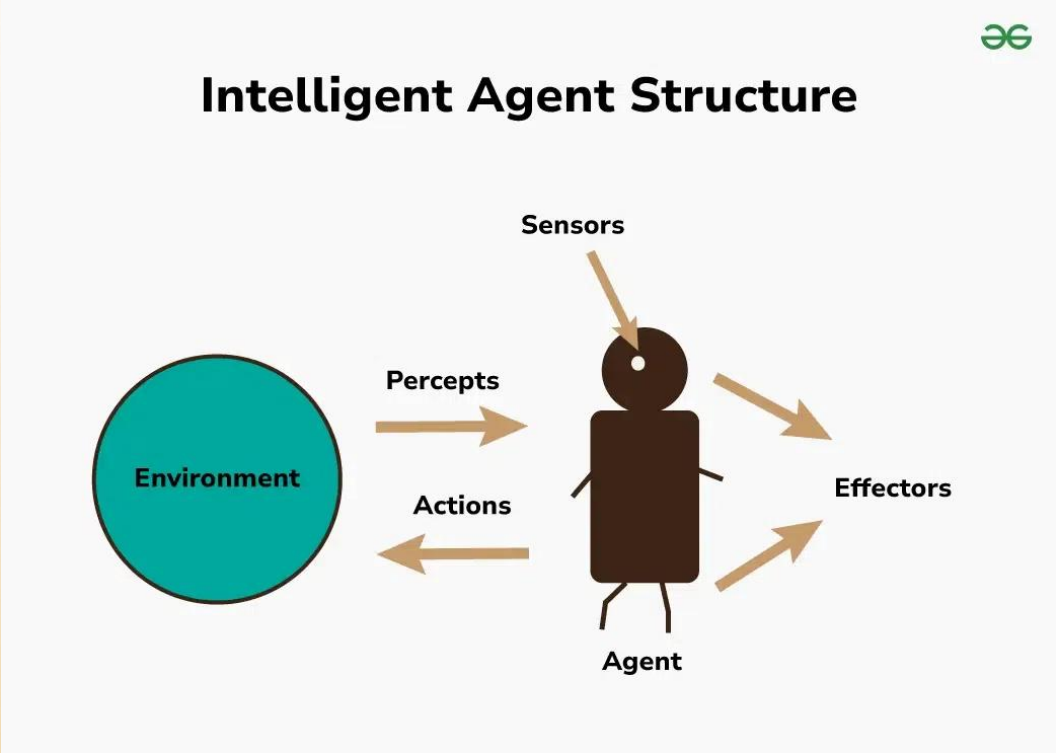
- Ancient roots: Aristotle (logic)
- 1936: Alan Turing – computation, Turing test
- **1956: Dartmouth conference – birth of AI**
- 1960s–70s: symbolic AI
- 1980s: expert systems
- 1990s: machine learning
- 2000s–now: deep learning, robotics, autonomous systems

Key Milestones In AI

- 1997: Deep Blue Beats Kasparov
- 2011: IBM Watson Wins Jeopardy
- 2016: Alphago Beats Lee Sedol
- 2022+: Generative AI (Chatgpt, DALL·E, Etc.)

Intelligent Agents

- Agent: perceives environment (sensors) & acts (actuators)
- Structure: sensors → percepts → agent function → actions → environment



Agent Architecture

- **Environment:** Environment is the area around the agent that it interacts with. An environment can be anything like A physical space, A room or A virtual space like A game world or the internet.

- **Sensors:** Sensors are tools that ai agent uses to perceive their environment. They can be any physical like cameras, microphones, temperature sensors or A software sensor that read data from files.

Agent Architecture

- **Actuators:** Actuators are tools that ai agent uses to interact with their environment through some actions. They can be any physical actuators like wheels, motors, robotic hands, or computer screens or they can be software actuators that send messages.
- **Effectors:** Effectors take instructions from decision making mechanism and translates them into actions and these actions are performed through actuators.

Types of Agents

1. Simple Reflex Agents (Thermostat)
2. Model-based Reflex Agents
3. Goal-based Agents
4. Utility-based Agents
5. Learning Agents

Wrap-up & Activity

👉 Next class: Types of Agents in details

- Activity: think of a real-life problem and identify
  - Sensors
  - Actions
  - Goals

- Example: Food delivery app → GPS, restaurant suggestion, minimize delivery time

Thank You

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